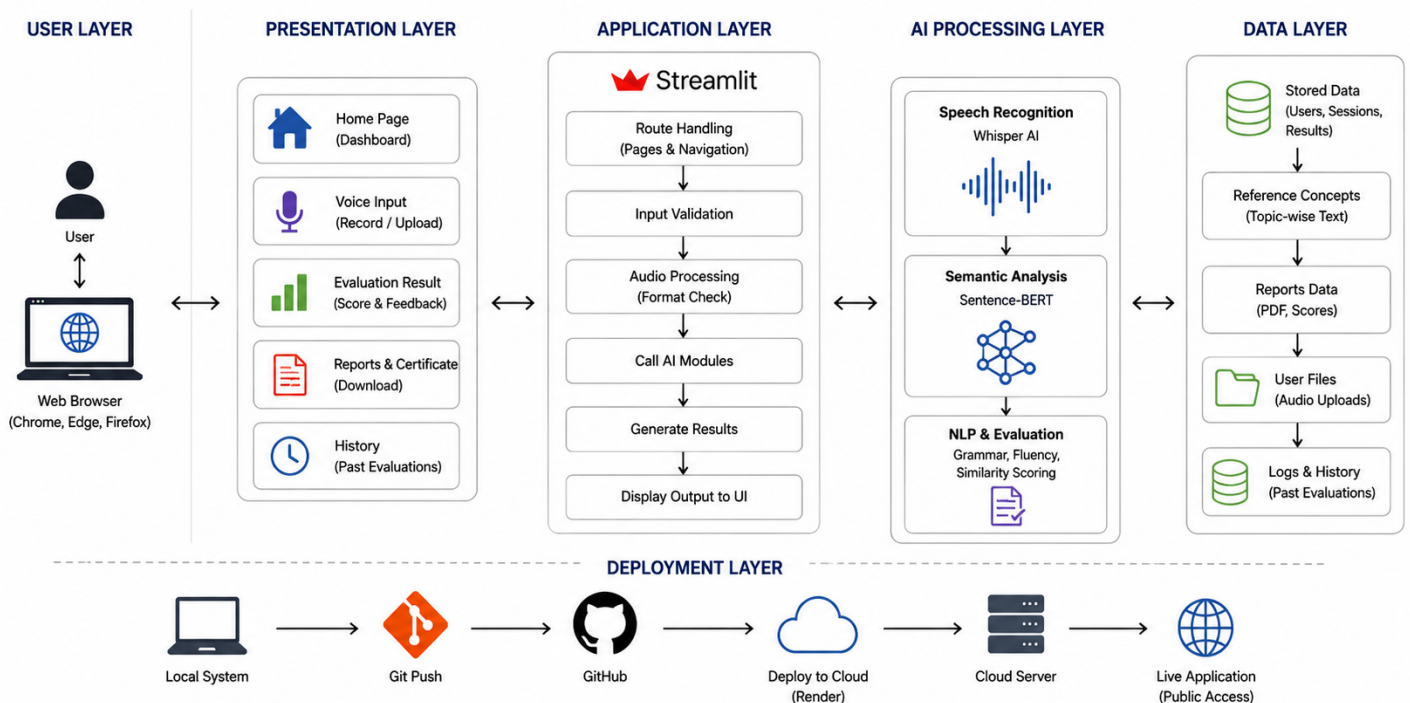


Solution Architecture

Date	18 July 2026
Team ID	XXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Voice-Based Concept Understanding Analyser – System Architecture



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	The client-side interface where students and educators interact with the application. Users can record or upload audio, view evaluation results, dashboards, reports, and certificates.	Streamlit, HTML5, CSS3, JavaScript
API Gateway / Load Balancer	Serves as the entry point for all incoming requests, managing traffic and securely routing it to the Streamlit application deployed on the cloud.	Nginx (Reverse Proxy) on Render
Auth Service	Handles user authentication, registration, and session management. Secures user routes and maintains encrypted user sessions.	Streamlit-Authenticator, JWT (JSON Web Tokens), Bcrypt
Core Logic Service	The central backend logic that processes audio inputs, performs speech-to-text conversion, evaluates semantic similarity, analyzes grammar, and generates AI feedback and scores.	Python, Whisper (AI), Sentence-BERT, NLTK / spaCy, Pandas, NumPy
External API Integrations	Integrates with external AI models and libraries for speech recognition and semantic analysis instead of building models from scratch.	OpenAI Whisper API (Local Model), Hugging Face Sentence-Transformers
Data / Storage Layer	Stores user data, audio files, transcripts, evaluation results, reports, and logs. Ensures secure, reliable, and scalable data management.	SQLite / PostgreSQL (Database), File System (Audio & Reports Storage)